



USER MANUAL LEGENDS FORD COUPE



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DEAR iRACING USER,

Congratulations on your iRacing membership, and we hope you enjoy the Legends Ford Coupe. From all of us at iRacing, we appreciate your support and your commitment to our product. We aim to deliver the ultimate sim racing experience, and we hope that you'll find plenty of excitement with us behind the wheel of your new car!

The following guide explains how to get the most out of your new car, from how to adjust its settings off of the track to what you'll see inside of the cockpit while driving. We hope that you'll find it useful in getting up to speed.

Thanks again for your continued support, and we'll see you on the track!



CHASSIS

DOUBLE WISHBONE INDEPENDENT
FRONT, LIVE AXLE 3-LINK REAR



LENGTH
3158mm
124in

WIDTH
1524mm
60in

WHEELBASE
1854mm
73in

DRY WEIGHT
476kg
1050lbs

WET WEIGHT
WITH DRIVER
571kg
1260lbs

POWER UNIT

NATURALLY ASPIRATED,
WATER COOLED, IN-LINE 3-CYLINDER



DISPLACEMENT
890cc
54CID

RPM LIMIT
12,100
RPM

TORQUE
76lb-ft
100Nm

POWER
130 bhp
97kW



INTRODUCTION

The information found in this guide is intended to provide a deeper understanding of the chassis setup adjustments available in the garage, so that you may use the garage to tune the chassis setup to your preference.

Before diving into chassis adjustments, though, it is best to become familiar with the car and track. To that end, we have provided baseline setups for each track commonly raced by these cars. To access the baseline setups, simply open the Garage, click iRacing Setups, and select the appropriate setup for your track of choice. If you are driving a track for which a dedicated baseline setup is not included, you may select a setup for a similar track to use as your baseline. After you have selected an appropriate setup, get on track

and focus on making smooth and consistent laps, identifying the proper racing line and experiencing tire wear and handling trends over a number of laps.

Once you are confident that you are nearing your driving potential with the included baseline setups, read on to begin tuning the car to your handling preferences.

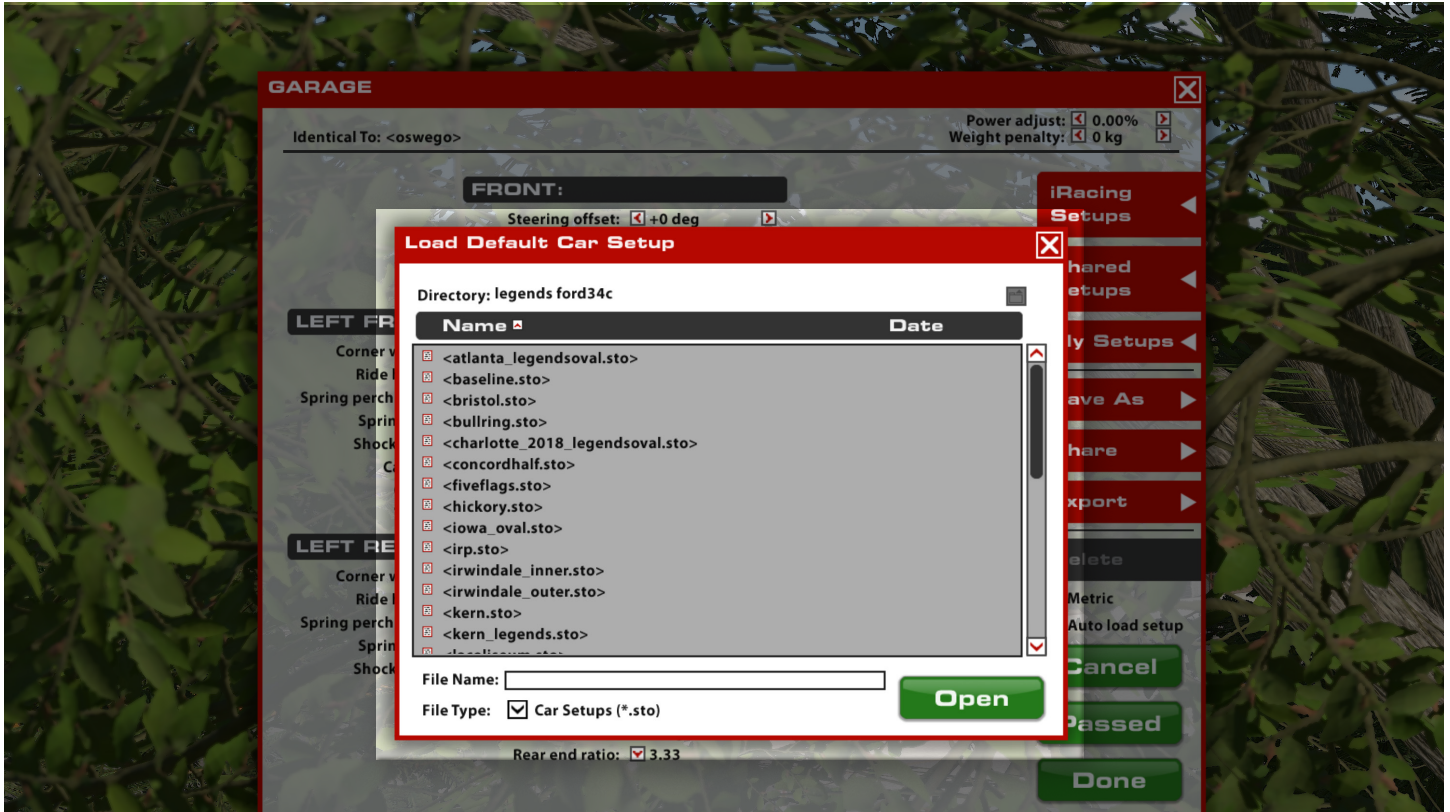
GETTING STARTED



Once you load into the car, select 1st gear. Slowly release the clutch while applying the throttle to drive away. A clutch is necessary when coming to a stop to prevent stalling the engine and shifting into reverse if necessary, but the clutch isn't required once the vehicle is in motion for upshifts or downshifts. To upshift, simply let off

the throttle and select the next higher gear. To downshift, give the throttle a blip while selecting the next lower gear. If you downshift too early, or don't blip the throttle sufficiently, the wheel speed and engine speed will be mismatched, leading to wheel hop at the rear and a possible spin.

LOADING AN iRACING SETUP



When you first load into a session, the iRacing Baseline setup will be automatically loaded onto the car. If you would like to try any of the other iRacing pre-built options, you may select it by going to Garage > iRacing Setups > and then selecting another option that fits your needs. If you would like to customize the setup, simply make the changes in the garage that you would like to update and click apply. If you would like to save your setup for future use click

“Save As” on the right to name and save the changes. To access all of your personally saved setups, click “My Setups” on the right side of the garage. If you would like to share a setup with another driver or everyone in a session, you can select “Share” on the right side of the garage to do so. If a driver is trying to share a setup with you, you will find it under “Shared Setups” on the right side of the garage as well.

DASH PAGES

A single-page AIM digital display conveys information to the driver in an easy-to-read format:



TACHOMETER BAR

Displays Engine RPM in a graphical segmented bar across the top of the display

RPM

Engine RPM

ECT

Displays the temperature, in °F or °C, of the engine cooling water

SPD

Car speed, in mph or kph

GEAR

Currently selected transmission gear

The display also features shift lights across the top, which will light up from left (green) to right (red) as RPM increases. When the optimum shift RPM is reached, a red LED will illuminate in the upper right corner of the display, shown below.



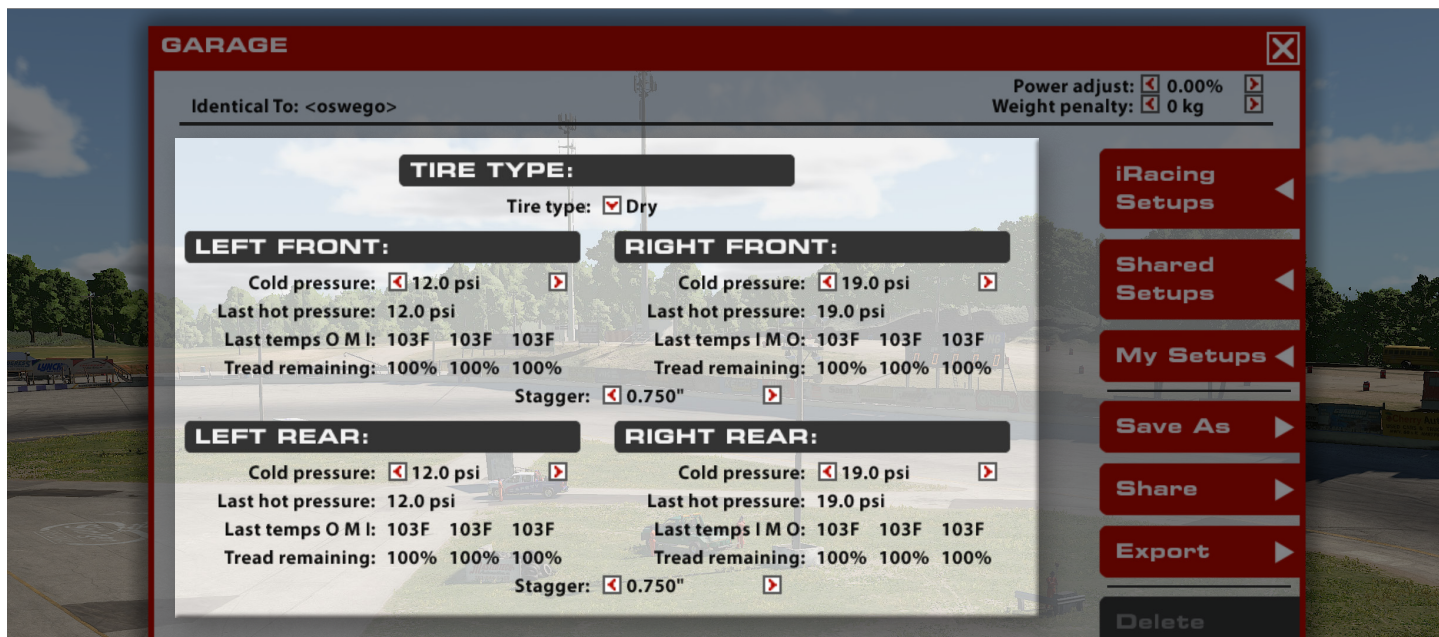
ADVANCED SETUP OPTIONS

This section is aimed toward more advanced users who want to dive deeper into the different aspects of the vehicle's setup. Making adjustments to the following parameters is not required and can lead to significant changes in the way a vehicle handles. It is recommended that any adjustments are made in an incremental fashion and only singular variables are adjusted before testing changes.



TIRE PAGE

TIRE SETTINGS (ALL FOUR TIRES)



TIRE TYPE

Selects which type of tire is installed on the car when loaded into the world. Dry, or slick, tires are used for dry racing conditions while Wet tires are intended for raining and wet track conditions.

COLD PRESSURE

Air pressure in the tire when the car is loaded into the world. Higher pressures will reduce heat buildup, but will provide better grip with higher loads and higher speeds. Lower pressures will increase heat buildup, but will provide better grip at lower loads and lower speeds. Cold pressures should be set to track characteristics for optimum performance.

LAST HOT PRESSURE

Air pressure in the tire after the car has returned to the pits. The difference between Cold and Hot pressures can be used to identify how the car is progressing through a run in terms of balance, with heavier-loaded tires seeing a larger difference between Cold and Hot pressures. On left-turn ovals, the right-front tire should always see the highest buildup on a balanced car, while the left sides should be roughly the same, but it is important to monitor the hot pressures after a run and adjust accordingly. Ideally, the difference in hot pressures on one side of the car should be roughly equal to the difference between cold pressures after a longer run.

LAST TEMPERATURES

Tire carcass temperatures once the car has returned from the pits. Wheel Loads and the amount of work a tire is doing on-track is reflected in the tire's temperature, and these values can be used to analyze the car's handling balance. Center temperatures are useful for directly comparing the work done by each tire, while the Inner and Outer temperatures are useful for analyzing the wheel alignment while on track. These values are measured in three zones across the tread of the tire.

TREAD REMAINING

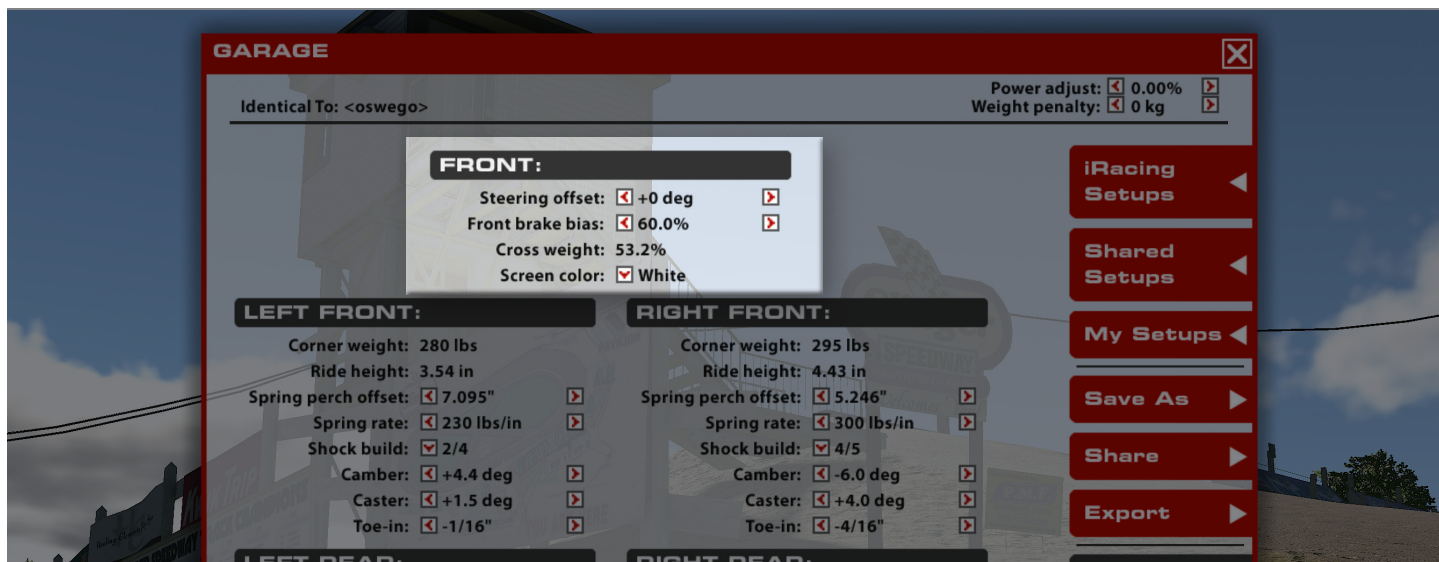
The amount of tread remaining on the tire once the car has returned from the pits. Tire wear is very helpful in identifying any possible issues with alignment, such as one side of the tire wearing excessively, and can be used in conjunction with tire temperatures to analyze the car's handling balance. These values are measured in three zones across the tread of the tire.

STAGGER

Stagger is the difference in circumference between the left- and right-side tires. For oval tracks, Stagger is necessary due to the differences in turning radii between the two tires and should always be greater than zero. Tracks with flatter turns or a tighter turn radius will need more stagger, while tracks with more banking and/or longer, wider turns will need less stagger. For road courses, keep Stagger at zero.

CHASSIS

FRONT END



STEERING OFFSET

Degrees of steering wheel offset, achieved with a combination of installing the steering wheel into the quick release mechanism off-center and adjusting front tie-rods. This can be used to compensate for chassis settings which place the wheel off center and is primarily a driver comfort adjustment.

FRONT BRAKE PRESSURE BIAS

Brake Bias is the percentage of braking force that is being sent to the front brakes. Values above 50% result in more pressure being sent to the front, while values less than 50% send more force to the rear. This should be tuned for both driver preference and track conditions to get the optimum braking performance for a given situation.

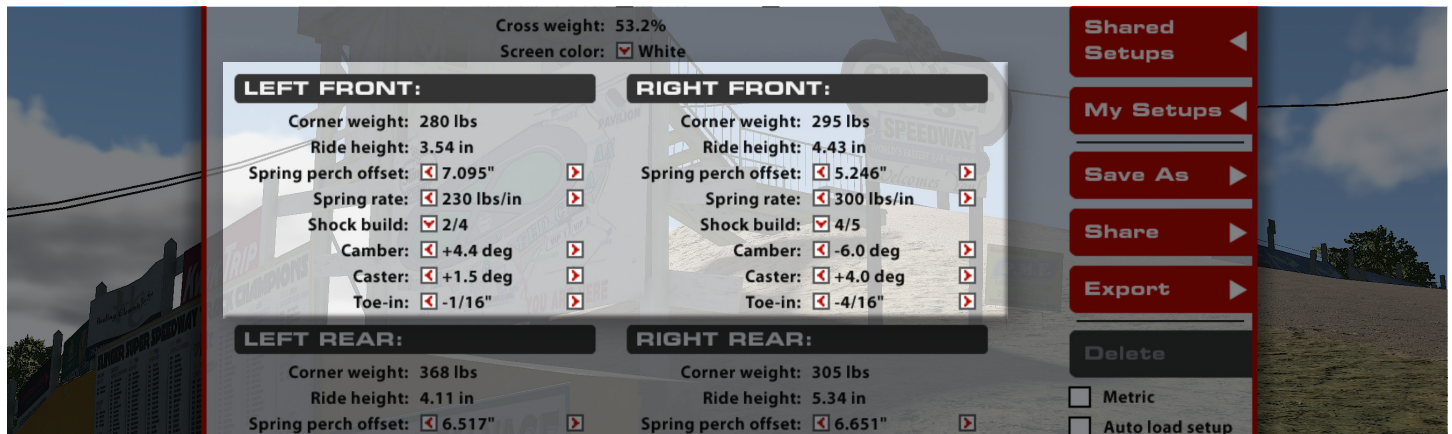
CROSS WEIGHT

Cross weight is the amount of weight on the car's Left-Rear and Right-Front tires relative to the entire weight of the car, displayed in percent. This is adjusted via the corner Shock Collar Offsets. For ovals, Crossweight is typically set a few percent higher or lower than 50% depending on the track's characteristics. For smaller ovals with tighter turns, a lower crossweight may be desired to help the car rotate through the corner. For higher speeds and larger turns, higher crossweight may be desired to aid in drive-off and maintain stability through the corner. For Road Courses, having the Crossweight right around 50% is usually the best place to start.

SCREEN COLOR

A choice of 7 different colors are available for the digital dash background: White, Cyan, Blue, Green, Yellow, Red and Purple. This option is adjustable from within the car as well as the garage.

FRONT CORNERS



CORNER WEIGHT

The weight underneath each tire under static conditions in the garage. Correct weight arrangement around the car is crucial for optimizing a car for a given track and conditions. Individual wheel weight adjustments and crossweight adjustments are made via the Spring Perch Offset setting.

RIDE HEIGHT

Distance from ground to a reference point on the chassis. Front heights are measured at the front of the lowest part of the frame just behind the front wheels. Adjusting Ride Heights is key for optimum performance, as they can directly influence the vehicle's mechanical grip as well as how much weight transfer occurs during braking and cornering.

SPRING PERCH OFFSET

Legend Car shocks feature a threaded collar as the upper spring perch mount. This collar is used to adjust ride height and corner weight, and adjusting this setting applies a preload to the spring under static conditions. Decreasing the value increases preload on the spring, adding weight to its corner and increasing the ride height at that corner. Increasing the value does the opposite, reducing height and weight on a given corner. These should be adjusted in pairs (left and right, for example) or in conjunction with all four spring perch adjustments in the car to prevent crossweight changes while adjusting ride height.

SPRING RATE

Spring Rate changes how stiff the spring is, represented in a force per unit of displacement. Primarily responsible for maintaining ride height and a good attitude under changing wheel loads, stiffer springs will maintain the car's attitude better while sacrificing mechanical grip. Softer springs will deal with bumps better and increase mechanical grip, but may cause the chassis to move around too much, hurting performance.

SHOCK BUILD

The Legend Car uses sealed, non-adjustable shocks. However, the suspension's damping behavior can be altered with different shock builds. Shocks are represented by two numbers representing the relative Compression and Rebound values, with higher numbers representing more force for that value. For example, the "4/2" and "4/5" shock both have the same Compression, but the "4/5" shock will have more Rebound. Stiffer shocks (higher values) will control the chassis's attitude better, producing a more responsive feel, but could hurt mechanical grip over rough surfaces. Softer shocks will be more compliant over bumps but could make the chassis feel unresponsive.

CAMBER

Camber is the vertical angle of the wheel relative to the center of the chassis. Negative camber places the top of the wheel closer to the chassis centerline than the bottom of the wheel, positive camber places the top of the tire farther out than the bottom. Greater camber angles will increase the cornering force generated by the tire, but will reduce the amount of longitudinal grip the tire will have under braking. For ovals, set the left side positive and the right side negative. For road courses, all four wheels should be set with negative camber.

CASTER

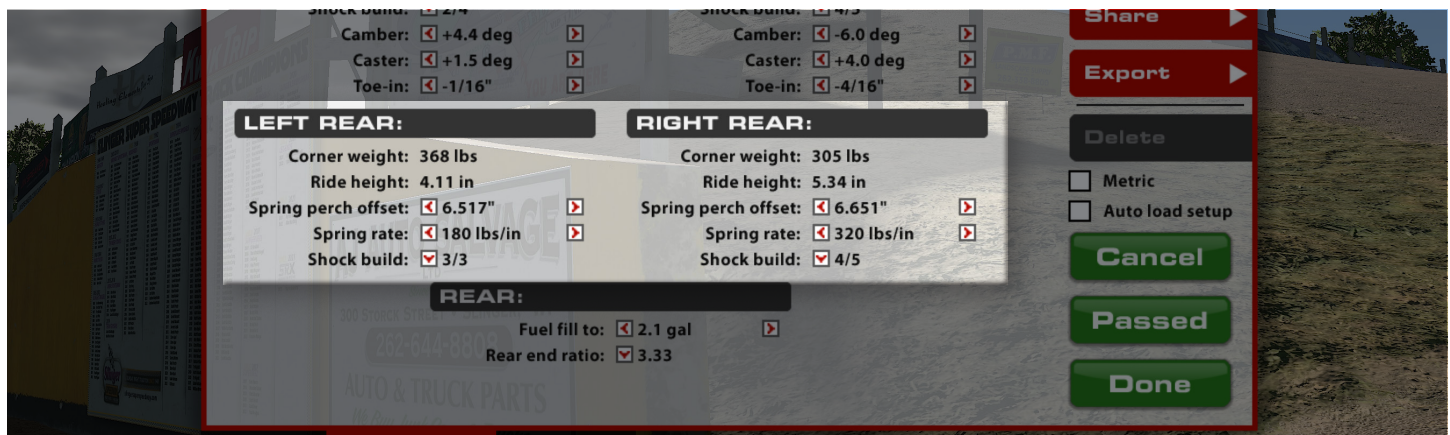
How much the steering axis is leaned back (positive) or forward (negative), which influences dynamic load jacking effects as the car is steered. More positive caster results in a heavier steering feel but decreases dynamic crossweight while turning, as well as adding straight-line stability. Running less caster on the left-front will cause the vehicle to pull to the left, a desirable effect on ovals.

TOE-IN

Toe is the angle of the wheel, when viewed from above, relative to the centerline of the chassis. Positive toe-in is when the front of the wheel is closer to the centerline than the rear of the wheel, and negative toe-in (toe-out) is when the front of the wheel is farther away from the centerline than the rear of the wheel. More net toe-

out will destabilize the car in a straight line and help with turn-in but increases slip angle on the front wheels and can risk over-slipping the front tires at high values, which can result in sudden understeer when cornering.

REAR CORNERS



CORNER WEIGHT

The weight underneath each tire under static conditions in the garage. Correct weight arrangement around the car is crucial for optimizing a car for a given track and conditions. Individual wheel weight adjustments and crossweight adjustments are made via the Spring Perch Offset setting.

RIDE HEIGHT

Distance from ground to a reference point on the chassis. Rear heights are measured at the rear of the lowest part of the frame just in front of the rear wheels. Adjusting Ride Heights is key for optimum performance, as they can directly influence the vehicle's mechanical grip as well as how much weight transfer occurs during braking and cornering.

SPRING PERCH OFFSET

Legend Car shocks feature a threaded collar as the upper spring perch mount. This collar is used to adjust ride height and corner weight, and adjusting this setting applies a preload to the spring under static conditions. Decreasing the value increases preload on the spring, adding weight to that corner and increasing the ride height at that corner. Increasing the value does the opposite, reducing height and weight on a given corner. These should be adjusted in pairs (left and right, for example) or in conjunction with all four spring perch adjustments in the car to prevent crossweight changes while adjusting ride height.

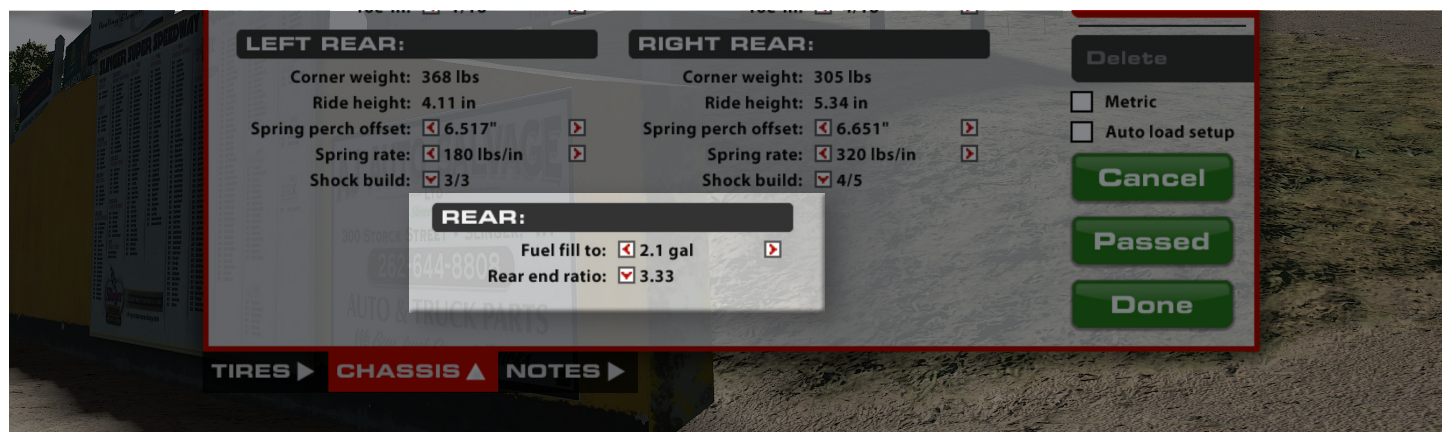
SPRING RATE

Spring Rate changes how stiff the spring is, represented in a force per unit of displacement. Primarily responsible for maintaining ride height and a good attitude under changing wheel loads, stiffer springs will maintain the car's attitude better while sacrificing mechanical grip. Softer springs will deal with bumps better and increase mechanical grip, may cause the chassis to move around too much, hurting performance.

SHOCK BUILD

The same sealed shocks used on the front of the car are also used at the rear. Shocks are represented by two numbers representing the relative Compression and Rebound values, with higher numbers representing more force for that value. For example, the "4/2" and "4/5" shock both have the same Compression, but the "4/5" shock will have more Rebound. Stiffer shocks (higher values) will control the chassis's attitude better, producing a more responsive feel, but could hurt mechanical grip over rough surfaces. Softer shocks will be more compliant over bumps but could make the chassis feel unresponsive.

REAR



FUEL FILL TO

This selects the amount of fuel in the fuel tank when the car leaves the garage.

REAR END RATIO

The Rear End Gear Ratio is the ratio between the driveshaft pinion and the differential ring gear. Higher number values produce better acceleration but reduce top speed, lower number values reduce acceleration but result in a higher top speed. Since Legend Cars feature a 5-speed transmission without changeable ratios, it's important to find the rear-end gear that works the best for each track. Because the transmission ratios are locked, not all ovals will see the best performance in 5th gear. Some will see lower lap times while running 4th gear with a lower Rear End ratio, so it's important to experiment with all possibilities.